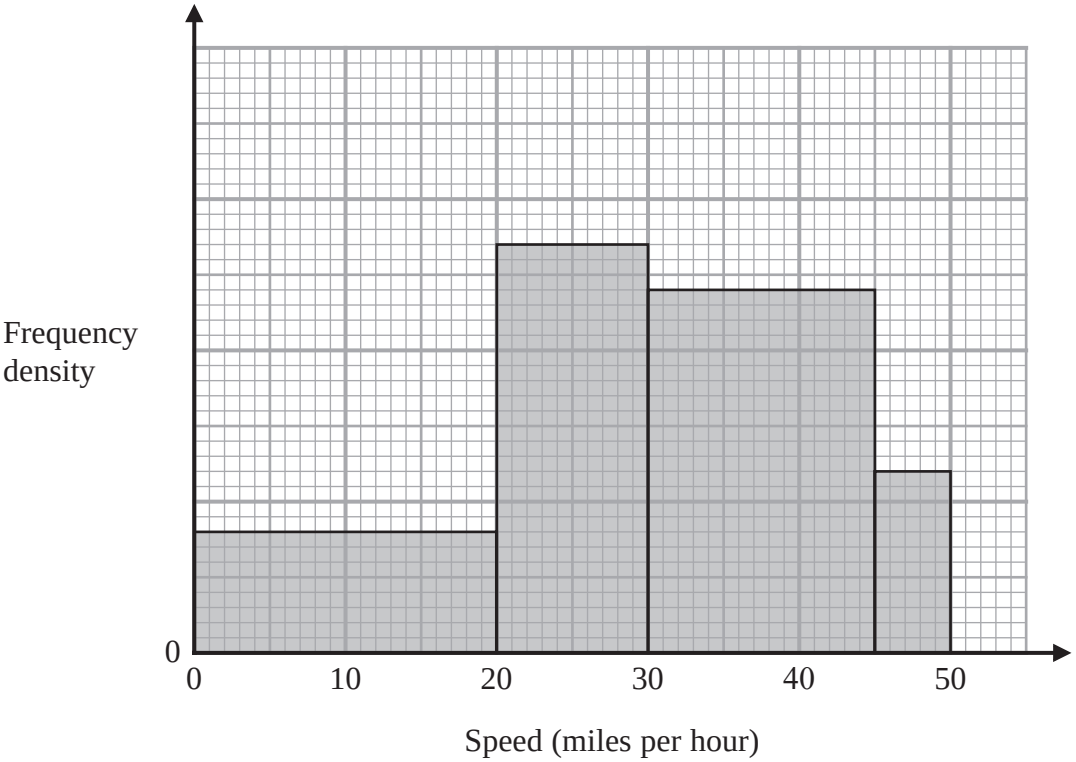
19 The histogram shows information about the speeds of cars travelling along a road on Monday.



16 of the cars travelled at a speed greater than 0 miles per hour but no more than 20 miles per hour.

None of the cars travelled at a speed greater than 50 miles per hour.

One of the cars is chosen at random.

(a) Find the probability that this car travelled at a speed greater than 20 miles per hour but no more than 30 miles per hour.

First density = 16 ÷ 20 = 0.8

Frequencies: 16, 10 × 2.7 = 27, 15 × 2.4 = 36,

5 × 1.2 = 6. Total = 85. P(20 < speed ≤ 30) = 27/85.

3 marks total
First bar fixes scale.
Use bar areas, then divide by all 85 cars.

27/85

(3)

(b) Work out an estimate for how many of the cars travelled at a speed greater than 40 miles per hour.

Above 40 mph: 5 × 2.4 + 5 × 1.2 = 12 + 6 = 18

2 marks total
Use only 40–45 and 45–50, not all of the 30–45 bar.

18

(2)

(Total for Question 19 is 5 marks)

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